

Nakuja Rocketry Club

Scientific Experiment Protocol & Rationale

Justification and Consolidated Test Protocol for the Scaled One-Grain
BATES Static Test: KNSB Propellant Characterization via
Thrust-Derived Ballistic Parameters

Solid Propulsion Team

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This document consolidates and supersedes the 29 June 2026 Justification for Conducting a Scaled One-Grain BATES Static Test and the associated Protocol: Determining Ballistic Parameters Without a Pressure Transducer, integrating both with the nozzle similarity analysis conducted in support of this test.

1 Introduction & Problem Statement

The Solid Propulsion Team is developing a 5-grain KNSB BATES motor for the N4 sub-orbital rocket. Two previous full-scale five-grain static tests resulted in casing rupture (CATO) between the third and fourth grain, with observed thrust exceeding simulation despite a simulated chamber pressure (4.4 MPa) well below the aluminum casing's burst pressure (12.2 MPa) — indicating unmodeled chamber pressure spikes rather than a simple overpressure event.

To de-risk this systematically, the team has adopted a four-stage static test campaign: a one-grain test, a two-grain test, a three-grain test, and a full five-grain test, each preserving bulkhead, nozzle, and casing dimensions from the final design. This document sets out the rationale, the theoretical grounding, and the measurement protocol for that campaign, with particular attention to two open technical questions raised during planning: the nozzle flow-separation warning returned by OpenMotor for the one-grain configuration, and how ballistic parameters are to be extracted from thrust data in the absence of a pressure transducer.

2 Background

2.1 Guidance from Richard Nakka and Prior N4 Experience

Richard Nakka strongly recommends BATES grains for propellant characterization, since they provide a relatively neutral thrust profile and a well-defined burning surface area, and notes that accurate burn rate coefficient (a) and pressure exponent (n) are best determined from static tests with controlled K_n ratios. He also warns that nozzle flow separation occurs in over-expanded nozzles when exit pressure is significantly lower than ambient, leading to unstable flow and inaccurate

performance predictions — a condition common when scaling down motor size while keeping nozzle throat diameter fixed.

Emmanuel Ogwen, an early N4 solid propulsion member, noted that the original purpose of the single-grain test was safety evaluation ahead of a full static fire and thrust curve profile analysis. That earlier effort encountered the same low-exit-pressure warning from OpenMotor and proceeded regardless, on the basis that the static test still provided critical grain-level data even with an unrepresentative nozzle exit condition. This document treats that same judgment call quantitatively rather than qualitatively, in Section 4.

3 Theoretical Background

3.1 The K_n Ratio and Chamber Pressure

Chamber pressure is governed by the ratio of burning surface area to throat area:

$$K_n = \frac{A_b}{A_t}, \quad P_c = (K_n a \rho_p c^*)^{\frac{1}{1-n}}$$

where a and n are the coefficients of Saint-Robert’s Law (Section 3.4), ρ_p is propellant density, and c^* is characteristic velocity. Because the throat area is fixed for a given nozzle, matching K_n between a sub-scale and full-scale motor reproduces the full-scale chamber pressure regardless of how the absolute burn area and throat area differ between the two. This relationship is a power law, not a linear one: doubling K_n does not double P_c except in the special case $n = 0$. Any comparison of chamber pressure across grain counts must go through this relationship explicitly rather than by direct proportion.

3.2 Nozzle Expansion Ratio and the Flow Separation Criterion

Exit pressure is set independently of P_c by the expansion ratio $\varepsilon = A_e/A_t$, via the isentropic area-Mach and pressure relations:

$$\varepsilon = \frac{1}{M_e} \left[\frac{2}{k+1} \left(1 + \frac{k-1}{2} M_e^2 \right) \right]^{\frac{k+1}{2(k-1)}}, \quad \frac{P_e}{P_c} = \left(1 + \frac{k-1}{2} M_e^2 \right)^{-\frac{k}{k-1}}$$

A nozzle is overexpanded when $P_e < P_a$ (ambient). Flow separation in the diverging section becomes likely once P_e/P_a falls below approximately 0.35–0.40 (Summerfield criterion); more severe overexpansion produces earlier and less predictable separation, and can introduce transient side loads on the motor mount. Because ε is fixed for a given nozzle, P_e/P_c is approximately constant through a burn, and whether separation occurs is governed by how P_c (via K_n) combines with ε against ambient pressure at the test site (≈ 81 – 85 kPa at JKUAT elevation, versus 101.3 kPa at sea level).

A sub-scale nozzle reproduces the full-scale design’s flow character only if *both* K_n and ε are matched simultaneously; matching K_n alone while leaving the exit unchanged increases ε and worsens overexpansion.

3.3 Thrust Coefficient and Its Dependence on Flow Attachment

The thrust coefficient C_f relates chamber pressure to thrust through nozzle geometry and exhaust gas properties:

$$F = C_f P_c A_t$$

Under fully attached, ideally-expanded or moderately over/under-expanded flow, C_f is well predicted from ε , k , and the pressure ratio P_c/P_a alone. Under separated flow, however, the effective flow no longer follows the full geometric expansion ratio — the flow detaches before the physical exit plane, and the effective C_f departs from the ideal isentropic value in a way that is not reliably predictable from ε and k alone. This has a direct consequence for Section 3.4: any thrust-based back-calculation of P_c that assumes an ideal C_f carries additional, difficult-to-bound error whenever the nozzle is operating in or near the separated regime. The severity of this error should be assessed per test stage using the P_e/P_a figures in Section 4, not assumed uniform across the campaign.

3.4 Deriving Ballistic Parameters from Thrust Alone

Where a pressure transducer is not available, a and n can be estimated from thrust-time data alone, subject to the C_f caveat of Section 3.3.

Data acquisition. Thrust $F(t)$ is recorded via load cell at a minimum sampling rate of 20 Hz, sufficient to resolve the transient ignition and tail-off phases.

Estimating chamber pressure. Chamber pressure is back-calculated from instantaneous thrust:

$$P_c = \frac{F}{A_t C_f}$$

where A_t is throat area (post-test throat measurement is required to correct for erosion over the burn) and C_f is estimated from expansion ratio and exhaust specific heat ratio as in Section 3.2 — valid only to the extent the flow is attached, per Section 3.3.

Calculating burn rate. Mass flow rate follows from P_c :

$$\dot{m} = \frac{P_c A_t}{c^*}, \quad \dot{m} = \rho_p A_b r \quad \Rightarrow \quad r = \frac{P_c A_t}{\rho_p A_b c^*}$$

where A_b is the instantaneous burning surface area, modeled from grain geometry and web regression over time.

Determining a and n . Saint-Robert's Law, $r = a P_c^n$, is linearized:

$$\log r = \log a + n \log P_c$$

Plotting $\log r$ against $\log P_c$ across the burn gives n as the slope and $\log a$ as the intercept. Accuracy depends on correctly modeling the time-varying A_b , and on the C_f assumption above; multiple tests should be averaged, and, per Section 5, treated as multiple points on the same underlying P_c - r curve across test stages rather than as independent single-test fits.

4 Current Configuration: Quantified Fixed-Nozzle Operating Point

New nozzle fabrication is not available for this test cycle; the stages fire through the existing 23 mm throat / 61 mm exit nozzle ($\varepsilon = 7.03$). Grain geometry (33.65–39 mm cores, 92 mm OD, 140 mm length per BATES segment) was used to reconstruct burning area per stage.

Quantity	1-grain	3-grain (bracket)	5-grain (design)
K_n	63.3	190–196	323–353
P_c (peak)	0.96 MPa [*]	2.6–5.1 MPa [†]	4.4–4.7 MPa
P_e (est.)	22.9 kPa	62–121 kPa [†]	97–111 kPa
P_e/P_a (local)	0.28	0.75–1.45 [†]	1.2–1.3
Flow regime	overexpanded, separated	near-attached to mildly underexpanded	attached

Table 1: Fixed-nozzle operating point by campaign stage. ^{*}Actual OpenMotor simulation result for the 1-grain, fixed-nozzle configuration, superseding earlier hand-estimates based on generic literature pressure exponents. [†]Range reflects uncertainty in the true pressure exponent n , which the campaign itself is designed to determine; both bounds place the 3-grain stage above the $P_e/P_a \approx 0.35$ –0.40 separation threshold.

Two conclusions follow. First, the 1-grain stage runs meaningfully overexpanded ($P_e/P_a \approx 0.28$) but at a chamber pressure (≈ 0.96 MPa) consistent with stable KNSB combustion, not the near-atmospheric, chuffing-risk regime implied by a naive literature- n extrapolation. Second, the 3-grain stage is very likely to run at or above the separation threshold on the existing nozzle regardless of where the true pressure exponent falls within its plausible range, meaning **no new nozzle hardware is expected to be required beyond the 1-grain stage**. This should be re-verified against the actual 1-grain firing data before the 3-grain test is finalized.

A single 1-grain firing constrains a and n from a Kn trace spanning a narrow window around $K_n \approx 63$, i.e. a narrow P_c window around ≈ 1 MPa; it does not by itself validate the fit across the full ≈ 1 –4.7 MPa design range. This is addressed structurally by the staged campaign (Section 5), not by this test alone.

5 Staged Test Campaign

5.1 Stage 1: One-Grain

Objectives: burn rate coefficient (a) and pressure exponent (n), burn regression behaviour, burn quality (void/microcrack evaluation across additive variants), ignition circuit efficiency, and igniter viability.

Caveats: C_f -based pressure back-calculation (Section 3.4) carries added uncertainty at this stage given $P_e/P_a \approx 0.28$; the a, n fit from this stage alone should be treated as a single, low- P_c anchor point rather than a design-validated result. Igniter energy sized and proven at this stage does not automatically carry forward to three or five grains, since total ignition surface area scales with grain count; igniter viability should be re-checked, not assumed, at each subsequent stage.

5.2 Stage 2: Two-Grain

Objectives: behavior of mass flow rate from two grains with varying core diameters, plausible pressure spikes emanating from direct scaling of grain number, plausible linearity in chamber pressure values (Both $P_{c_{peak}}$ and $P_{c_{average}}$ with increase in grain number) and justification of ignition circuit viability.

5.3 Stage 4: Three-Grain

Objectives: effects of grain stacking, with particular attention to flow conditions at the trailing (nozzle-end) grain. This is the first stage at which gas from upstream grains flows through the port of a downstream grain before reaching the throat — an erosive-burning and mass-flux effect that is structurally absent from an isolated single-grain test and cannot be recovered from Stage 1 data by any correction. It is also, per Section 4, expected to sit above the flow-separation threshold on the existing nozzle, making thrust-curve and C_f -based data from this stage substantially more trustworthy than Stage 1's.

5.4 Stage 3: Five-Grain

Objectives: flight viability and full systems check ahead of launch, at the design operating point.

5.5 Multi-Point Ballistic Calibration Across Stages

Each stage should be treated as an additional calibration point on the same underlying $\log r$ vs. $\log P_c$ relationship (Section 3.4), not solely as a hardware milestone. Stage 1 anchors the curve near $P_c \approx 1$ MPa; Stage 2 anchors it near $P_c \approx 3$ –5 MPa; Stage 3 confirms it at the design point. Fitting a and n across all three, rather than from Stage 1 in isolation, is the direct mitigation for the narrow-window limitation noted above.

6 Experimental Design and Methods

- Produce reference grains using current best practices (pressure curing, fine-milled KNO_3).
- Section sacrificial grains for microscopic void/microcrack analysis.
- Instrument every static test with a load cell sampling at ≥ 20 Hz, sufficient to resolve ignition transients and tail-off.
- Measure post-test throat diameter on every firing to correct A_t for erosion before it enters the P_c back-calculation of Section 3.4.
- Test additive variants (surfactants, plasticizers) under realistic motor geometry to evaluate void/microcrack reduction and pressure-spike mitigation.
- Compare measured thrust curves and derived burn rate against OpenMotor predictions at each stage.

7 Success Criteria

- Stable pressure/thrust curve without extreme, unmodeled spikes.
- Burn rate data consistent enough, across all three stages, to support a single a, n fit applicable across the full design P_c range.
- Microscopic evidence of reduced defects in additive variants.
- Igniter reliability confirmed independently at each grain count.

8 Risk Mitigation and Next Steps

- Perform a dry run before each live firing.
- Re-verify the Section 4 operating-point estimates against actual 1-grain firing data before finalizing the 3-grain nozzle configuration.
- If a throat insert becomes feasible (material constraints permitting), a small graphite, phenolic, or ablative-composite restrictor sized toward the design K_n remains a lower-cost alternative to new nozzle fabrication, and should be revisited if Stage 1 data suggests the 3-grain bracket in Section 4 is optimistic.
- Document all deviations from predicted behaviour clearly at each stage, including the resolved P_e/P_a once measured, for use in the next stage's planning.
- Proceed to the next campaign stage only once the current stage's success criteria are met.

9 Conclusion

This methodical, geometry-preserving, staged approach follows best practices in amateur rocketry development. The nozzle flow-separation condition flagged by OpenMotor for the one-grain configuration is real but bounded: it is most severe at Stage 1, quantifiably resolves by Stage 2 without requiring new hardware, and does not compromise the campaign's core ballistic-characterization objective provided a and n are fit across all three stages rather than from Stage 1 in isolation. This positions the team for a successful full-scale motor while keeping structural risk low throughout, consistent with the team's existing failure-analysis practice.

References

1. Solid Propulsion Team, *Justification for Conducting a Scaled One-Grain BATES Static Test for KNSB Propellant Characterization in Support of 5-Grain Solid Rocket Motor Development*, 29 June 2026.
2. Solid Propulsion Team, *Protocol: Determining Ballistic Parameters Without a Pressure Transducer*.
3. OpenMotor simulation output, `N4_newSRM.ric`: five-grain and one-grain (fixed-nozzle) motor statistics, K_n /pressure/thrust and nozzle exit pressure traces.

4. R. Nakka, *Amateur Rocket Motor Internal Ballistics* and nozzle theory notes, for the isentropic flow relations and overexpansion/separation criterion referenced in Section 3.